Low-Data RGB-to-NIR Translation with Augmentation and Pseudo-Label Learning

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Abstract

Near-infrared (NIR) imagery plays an important role in agricultural and remote sensing applications such as vegetation monitoring and environmental analysis. The multispectral sensors required to acquire it carry higher cost and integration overhead than standard RGB cameras, which motivates methods that infer NIR information directly from RGB imagery. Existing RGB-to-NIR translation approaches focus primarily on architecture and methodology, but their performance is constrained by the scarcity of paired training data, and the question of how to achieve reliable results when paired data is limited has received little direct attention. This work explores two complementary strategies for addressing this constraint: data augmentation and pseudo-label semi-supervised learning. Using the Canola UAV dataset and the SEN12MS satellite dataset, two architectures of substantially different capacity, U-Lite and MIRNet-v2, were trained under five augmentation configurations and a semi-supervised pipeline that treats model predictions on unlabelled RGB images as pseudo-labels. Augmentation consistently improved performance across both datasets and architectures, with the best supervised result reaching 29.74 dB PSNR on Canola. Semi-supervised learning produced smaller gains, improving performance on Canola when combined with augmentation but failing to improve over the augmented baseline on SEN12MS. The findings indicate that augmentation is the more dependable of the two strategies in the low-data regime, while the effectiveness of pseudo-labelling depends on whether the supervised baseline has learned the target dataset well enough for its predictions to act as useful training signal.

Keywords: Infrared Imaging, RGB-to-NIR, Image Translation, Image Augmentation

1 Introduction

Near-infrared (NIR) imagery plays an important role in remote sensing applications such as vegetation monitoring and environmental analysis [Gutman et al., 2021, Vescovo et al., 2012, Badgley et al., 2017]. Unlike visible red-green-blue (RGB) imagery, NIR captures material and structural properties that are not directly perceived in the visible spectrum, including indicators of plant health and surface reflectance [Gutman et al., 2021, Badgley et al., 2017]. However, dedicated NIR and multispectral sensors carry higher cost and integration overhead than standard RGB cameras, and are often unavailable on consumer-grade or resource-constrained platforms. This motivates methods that infer NIR information directly from widely available RGB imagery, enabling NIR-like benefits without additional hardware [Aslahishahri et al., 2021, Picon et al., 2022, Jin et al., 2025].

RGB-to-NIR translation can be framed as an image-to-image regression problem, in which a model learns a mapping from visible bands to a continuous NIR channel [Aslahishahri et al., 2021, Isola et al., 2017]. Recent work in deep learning, particularly using convolutional and encoder-decoder architectures, has shown strong performance on related image translation tasks. Learning an accurate RGB-to-NIR mapping nonetheless remains challenging. The relationship between visible and near-infrared reflectance is highly nonlinear and scene-dependent, and NIR labels typically require specialised sensors and careful calibration [Aslahishahri et al., 2021].

Available RGB-NIR datasets are often limited in size, geographic coverage, or diversity of land cover conditions [Aslahishahri et al., 2021].

Two limitations follow from these constraints. Models trained on small, homogeneous datasets tend to overfit and fail to generalise to new locations, seasons, or sensor configurations [Aslahishahri et al., 2021, Bansal et al., 2022]. Higher-capacity architectures that could exploit richer context are difficult to apply effectively in this regime, since they typically require large quantities of annotated data to avoid poor generalisation [Aslahishahri et al., 2021, Jin et al., 2025].

Image augmentation is a widely adopted strategy for mitigating data scarcity in deep learning. It increases the effective size and diversity of a dataset by applying label-preserving geometric and photometric transformations to existing samples, enabling models to generalise better from limited data [Kumar et al., 2024]. Complementarily, pseudo-labelling provides a semi-supervised learning mechanism that leverages a small set of annotated data to generate approximate labels for unlabelled samples [van Engelen and Hoos, 2020]. In this approach, a model trained on labelled data is used to infer targets for unseen inputs, which are then incorporated into the training process to improve performance and generalisation [Su et al., 2024]. While both techniques have demonstrated effectiveness across various vision tasks, their impact on RGB-to-NIR translation remains underexplored. This is particularly relevant given the scarcity of paired RGB–NIR datasets, where such methods could play a crucial role in improving model robustness and scalability.

Motivated by these challenges, this work investigates two complementary strategies for RGB-to-NIR translation under data-scarce conditions: data augmentation and pseudo-label-based semi-supervised learning. Augmentation expands the effective diversity of the labelled training set through geometric and photometric transformations applied to RGB-NIR pairs. Semi-supervised learning extends the training signal to unlabelled RGB images by treating predictions from a supervised baseline as approximate ground truth labels. Both strategies are evaluated on two architectures of substantially different capacity, U-Lite and MIRNet-v2, and across two remote sensing datasets, the canola subset of the Aslahishahri et al. [Aslahishahri et al., 2021] aerial dataset and the SEN12MS satellite dataset [Schmitt et al., 2019]. The combination allows the experiments to test whether observed gains hold across architectures and across remote sensing domains. All codes and datasets are available at: `https://github.com/yearat/RGB2NIR-Low-Data-Augmentation`

2 Related Works

Image-to-image (I2I) translation describes the class of problems in which a model learns to convert one representation of a scene into another, for instance from an edge map to a photograph, or from a visible-spectrum image to a synthetic near-infrared output, using a unified learning framework rather than task-specific engineering [Isola et al., 2017]. Rather than generating entirely new content, I2I methods condition their output on an observed input, enabling applications such as style transfer, image colourisation, modality conversion, and semantic segmentation [Isola et al., 2017, Minaee et al., 2021].

In remote sensing, I2I translation is relevant because aerial and satellite imagery is multi-spectral, with different spectral bands capturing complementary physical properties of the observed scene [Aslahishahri et al., 2021, Gutman et al., 2021]. Near-infrared (NIR) imagery encodes information related to vegetation health and surface reflectance that is not directly observable in the visible spectrum [Gutman et al., 2021, Badgley et al., 2017, Vescovo et al., 2012]. Translating between spectral domains therefore has practical value in precision agriculture, environmental monitoring, and land-use analysis [Gutman et al., 2021, Badgley et al., 2017]and synthesising NIR from RGB imagery reduces the instrumentation burden of multispectral data collection [Aslahishahri et al., 2021].

Due to the cost and complexity of dedicated NIR imaging systems, alongside the widespread availability of RGB cameras, recent research has increasingly focused on deep learning-based RGB-to-NIR translation. These approaches leverage image-to-image (I2I) models to synthesise NIR imagery directly from RGB inputs [Illarionova et al., 2021, Aslahishahri et al., 2021, Hossain et al., 2025]. Given that vegetation health indicators are primarily derived from NIR data, many studies utilise agricultural datasets and evaluate the effectiveness of generated NIR images through downstream metrics such as the Normalized Difference Vegetation Index (NDVI) and the Normalized Difference Red Edge (NDRE) [Zhao et al., 2024, Davidson et al., 2022,

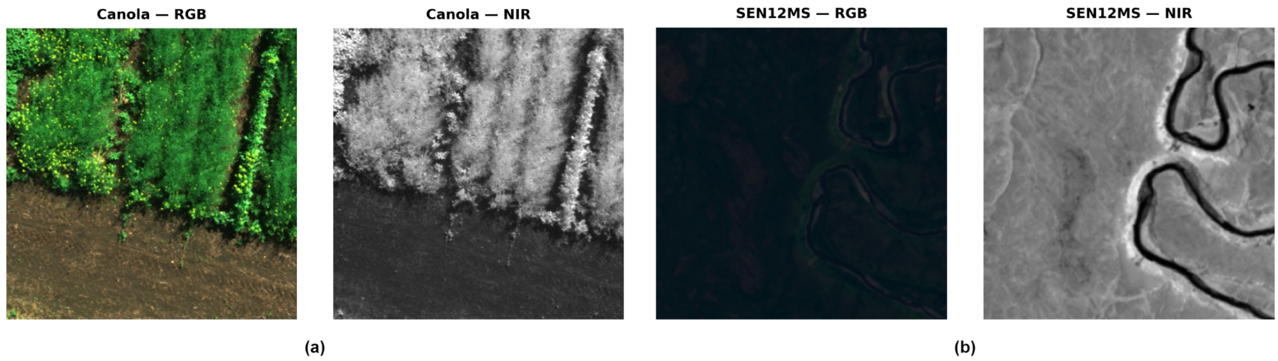

Figure 1: Representative paired RGB–NIR samples from the datasets used in this study. (a) Canola dataset showing UAV aerial imagery of crop canopy. (b) SEN12MS dataset showing Sentinel-2 satellite imagery. In both cases, the left image represents the RGB input and the right image represents the corresponding NIR target.

Aslahishahri et al., 2021]. Despite their effectiveness, these methods typically rely on pixel-wise paired RGB–NIR datasets, the collection of which is both technically demanding and resource-intensive. Such datasets require specialised imaging systems, careful sensor alignment, and radiometric calibration to ensure consistency and usability [Aslahishahri et al., 2021, Hossain and Dev, 2026].

Deep learning models generally achieve improved accuracy and generalisation when trained on large-scale datasets [Krizhevsky et al., 2012]. However, obtaining sufficient labelled data remains a significant challenge in many domains, including RGB-to-NIR translation. Addressing data scarcity is therefore an active area of research in machine learning [Alzubaidi et al., 2023]. One widely adopted approach is data augmentation, which involves generating additional training samples through label-preserving transformations such as geometric and photometric variations [Kumar et al., 2024]. Image augmentation has proven effective in improving model robustness and generalisation, particularly in low-data regimes [Rebuffi et al., 2021, Xu et al., 2023].

Another promising strategy is pseudo-label-based semi-supervised learning. In this approach, a model is first trained on a limited set of labelled data and then used to generate approximate labels for unlabelled samples [van Engelen and Hoos, 2020, Alzubaidi et al., 2023]. These pseudo-labelled data are subsequently incorporated into the training process, enabling the model to learn from a larger effective dataset and achieve improved performance and generalisation [Su et al., 2024].

Despite the data-intensive nature of RGB-to-NIR translation and the challenges associated with dataset collection, existing work has largely overlooked the problem of improving performance under limited data conditions. To address this gap, this work investigates the effectiveness of image augmentation and pseudo-label-based semi-supervised learning for enhancing RGB-to-NIR translation in data-scarce scenarios.

3 Methodology

3.1 Datasets

Two datasets are used in this work: a UAV-based agricultural dataset and a satellite remote sensing dataset, enabling evaluation of the proposed methods across different domains and testing their robustness and generalisation capabilities under varying imaging conditions and scene characteristics. Representative RGB–NIR image pairs from both datasets are shown in Figure 1.

The first dataset is the canola subset of the paired RGB–NIR aerial image dataset introduced by Aslahishahri et al. [Aslahishahri et al., 2021], captured using a UAV-mounted multispectral sensor over agricultural fields in Canada. The second dataset is derived from the SEN12MS dataset proposed by Schmitt et al. [Schmitt et al., 2019], using Sentinel-2 satellite imagery. For both datasets, 360 paired RGB–NIR samples are used in this work, with 300 samples used for training and 60 reserved for testing. All images are cropped to 256×256 patches prior to training. While the Canola dataset contains UAV-based crop imagery, SEN12MS provides large-scale satellite

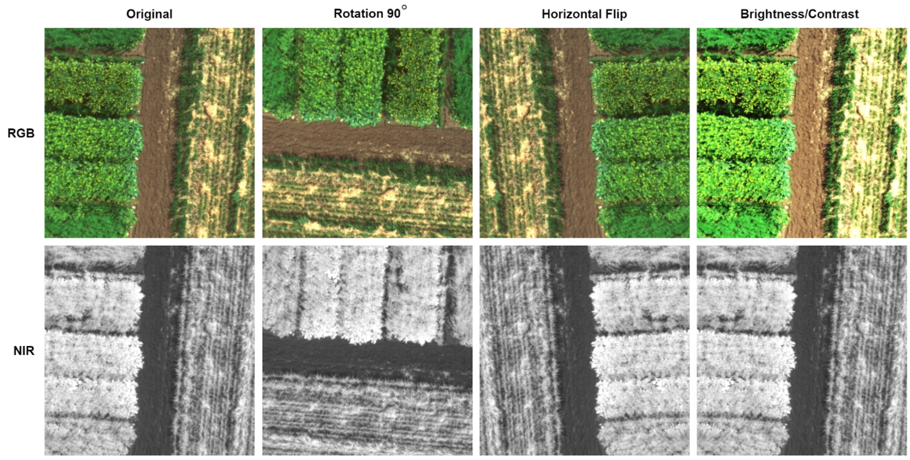

Figure 2: Examples of augmentation methods used in this study. From left to right: original image, 90° rotation, horizontal flip, brightness and contrast adjustment. Geometric augmentations are applied pairwise to both the RGB input and corresponding NIR target, while photometric augmentations are applied only to the RGB input and the NIR target remains unchanged.

imagery across diverse land-cover conditions, enabling evaluation across substantially different remote sensing domains.

3.2 Image Augmentation Pipeline

Five augmentation configurations are evaluated to analyse the effect of different augmentation strategies under limited-data conditions. The applied transformations include horizontal flip, vertical flip, rotations of 90°, 180°, and 270°, brightness jitter, and contrast jitter, with representative examples shown in Figure 2. Brightness and contrast augmentations are applied only to the input RGB images while the target NIR images remain unchanged, whereas geometric transformations are applied consistently to both RGB and NIR pairs to preserve spatial alignment. Each configuration is trained and evaluated independently using the same 300 training images, and all augmented samples are generated prior to training and kept fixed across epochs to ensure reproducibility.

3.3 Semi-Supervised Learning Pipeline

For the semi-supervised learning (SSL) experiments, the original 300 training pairs are divided into 200 labelled RGB–NIR pairs and 100 unlabelled RGB images, while the remaining 60 paired samples are reserved for evaluation and checkpoint selection. The split is generated once using a fixed random seed and kept consistent across all SSL experiments. A supervised baseline model is first trained on the 200 labelled pairs and then used to generate pseudo-labelled NIR outputs for the 100 unlabelled RGB images. Pseudo-labels are generated from the original unaugmented RGB inputs without confidence filtering, and all pseudo-labelled samples are retained for training. Two SSL settings are evaluated: without augmentation and with augmentation. In the no-augmentation setting, the mixed dataset contains 200 real labelled pairs and 100 pseudo-labelled pairs, giving 300 training samples in total. In the augmentation setting, all seven augmentation transforms are applied, expanding the 200 labelled pairs to 1,600 samples and the 100 pseudo-labelled pairs to 800 samples, resulting in 2,400 training samples overall. For each setting, two incorporation strategies are investigated: retraining a new model from scratch on the mixed dataset for 150 epochs using a learning rate of 2×10^{-4}, and fine-tuning from the baseline weights for 50 epochs using a reduced learning rate of 2×10^{-5}.

3.4 Models

Two architectures of substantially different capacity are evaluated. The first is a modified U-Lite [Dinh et al., 2023], a lightweight U-Net-based architecture using axial depthwise convolutions for computational efficiency. U-Net is chosen as it is one of the most widely used image-to-image (I2I) models and forms the basis of many subsequent architectures. The channel configuration is widened from [16, 32, 64, 128, 256, 512] to [20, 40, 80, 160, 320, 640], resulting in approximately 1.36 million trainable parameters. The second model is MIRNet-v2 [Zamir et al., 2023], a higher-capacity multi-scale restoration architecture that has achieved state-of-the-art performance across various image restoration tasks. Both models are trained using a reconstruction-only objective without adversarial learning, which is often unstable under limited paired data conditions [Goodfellow et al., 2014].

3.5 Implementation Details

All experiments are conducted on an NVIDIA GeForce RTX 3060 12GB GPU using mixed precision training to reduce memory usage and accelerate training. A global random seed of 42 is used across all experiments to ensure reproducibility. Models are trained using the Adam optimiser with a learning rate of 2×10^{-4}, a batch size of 4, and a CosineAnnealingLR scheduler over 150 epochs. Gradient clipping with a maximum norm of 1.0 is applied to improve training stability. Model performance is evaluated using Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM), two widely used reference-based image quality assessment metrics for evaluating image reconstruction and restoration quality.

4 Results

4.1 Supervised Learning with Augmentation

Table 1 presents PSNR and SSIM results for both models across five augmentation strategies on Canola and SEN12MS datasets.

Model	**Augmentation**	**Canola**		**SEN12MS**	
		PSNR	**SSIM**	**PSNR**	**SSIM**
MIRNet-v2	Baseline	28.62	0.9280	28.14	0.8245
	Flips	29.49	0.9394	**29.83**	**0.8495**
	Rotations	**29.74**	**0.9414**	29.28	0.8435
	Brightness	29.44	0.9372	28.99	0.8380
	Contrast	29.44	0.9372	28.99	0.8380
U-Lite	Baseline	27.95	0.9103	27.10	0.7728
	Flips	28.34	0.9222	27.92	0.8048
	Rotations	**28.36**	**0.9230**	**27.96**	**0.8093**
	Brightness	28.30	0.9200	27.73	0.7977
	Contrast	28.30	0.9200	27.73	0.7977

Table 1: Supervised learning results across augmentation strategies and datasets.

All augmentation strategies improve upon the no-augmentation baseline for both models and datasets. Geometric augmentations (flips and rotations) yield larger gains than photometric jitter: on Canola, rotations achieve 29.74 dB (MIRNet-v2, +1.12 dB), while brightness and contrast both reach 29.44 dB (+0.82 dB). A similar pattern appears on SEN12MS: flips yield 29.83 dB (+1.69 dB) while photometric augmentation yields 28.99 dB (+0.85 dB). MIRNet-v2 outperforms U-Lite across all conditions, with the gap widening on SEN12MS (1.87 dB at best) compared to Canola (1.38 dB).

4.2 Semi-Supervised Learning Results

Table 2 reports PSNR and SSIM for both models across all SSL conditions on both datasets.

Model	Aug.	SSL	Canola		SEN12MS	
			PSNR	SSIM	PSNR	SSIM
MIRNet-v2	None	Baseline	28.23	0.9235	27.39	0.8099
	None	Retrain	28.15	0.9244	27.62	0.8151
	None	Finetune	28.26	0.9239	27.41	0.8106
	Aug all	Baseline	29.42	0.9392	**29.71**	0.8492
	Aug all	Retrain	**29.54**	**0.9418**	28.65	0.8431
	Aug all	Finetune	29.53	0.9397	29.42	**0.8493**
U-Lite	None	Baseline	27.46	0.9013	26.50	0.7616
	None	Retrain	27.65	0.9056	26.80	0.7682
	None	Finetune	27.45	0.9016	26.51	0.7616
	Aug all	Baseline	27.87	**0.9228**	27.49	0.8043
	Aug all	Retrain	**28.27**	0.9181	26.95	**0.8109**
	Aug all	Finetune	27.97	0.9129	**27.58**	0.8067

Table 2: Semi-supervised learning results. PSNR is reported in dB. Best PSNR and SSIM values for each model-dataset combination are highlighted in **bold**.

In the no-augmentation SSL condition, pseudo-labelling produces only marginal changes, with small gains in some settings and negligible decreases in others. MIRNet-v2 on Canola improves from 28.23 dB (baseline) to 28.26 dB (finetune, +0.03 dB); on SEN12MS, retrain reaches 27.62 dB (+0.23 dB). U-Lite shows a similar pattern, with retrain gains of +0.19 dB on Canola and +0.30 dB on SEN12MS. Augmented SSL yields larger gains on Canola: MIRNet-v2 achieves 29.54 dB (retrain, +1.31 dB from no-augmentation baseline of 28.23 dB), and U-Lite achieves 28.27 dB (retrain, +0.81 dB). However, on SEN12MS, augmented SSL does not consistently improve over the augmented baseline. MIRNet-v2 reaches 29.71 dB at the augmented baseline but drops to 28.65 dB under retrain and 29.42 dB under finetune. U-Lite exhibits the same pattern, with retrain (26.95 dB) falling below the augmented baseline (27.49 dB), while finetune (27.58 dB) slightly exceeds it.

5 Discussion

The results demonstrate that data augmentation consistently improves RGB-to-NIR translation performance across both architectures and datasets, with geometric augmentations providing the strongest gains. MIRNet-v2 consistently outperforms the lightweight U-Lite model, particularly on the more complex SEN12MS dataset. Semi-supervised learning without augmentation provides only marginal improvements, suggesting that pseudo-label quality is limited when trained on small labelled datasets. Since pseudo-labels are generated by the baseline model itself, prediction errors can be propagated back into training, limiting the amount of new information introduced by the additional data. When combined with augmentation, SSL improves performance on the more homogeneous Canola dataset, but produces mixed results on the more diverse SEN12MS dataset, where the greater variability of the data likely increases pseudo-label noise, reducing the effectiveness of retraining. Overall, the findings indicate that augmentation is a reliable strategy for improving RGB-to-NIR translation under limited-data conditions, while the effectiveness of pseudo-label-based SSL depends strongly on dataset complexity and pseudo-label quality.

6 Conclusion

This work investigated data augmentation and pseudo-label-based semi-supervised learning for RGB-to-NIR translation under limited-data conditions across two architectures and two remote sensing datasets. The results show that data augmentation consistently improves performance, particularly with geometric transformations, while the effectiveness of pseudo-label-based SSL depends strongly on dataset complexity and pseudo-label quality. Overall, augmentation proved to be a reliable strategy for low-data RGB-to-NIR translation, whereas SSL produced mixed results on more diverse datasets. Future work will explore more advanced SSL methods and evaluate the generated NIR imagery on downstream remote sensing tasks such as NDVI analysis.

Acknowledgments

This research was conducted with the financial support of Research Ireland under Grant Agreement No. 13/RC/2106_P2 at the ADAPT Centre at Trinity College Dublin. ADAPT, the Research Ireland Centre for AI-Driven Digital Content Technology, is funded through the Research Ireland Centres Programme.

References

[Alzubaidi et al., 2023] Alzubaidi, L., Bai, J., Al-Sabaawi, A., Santamaría, J., Albahri, A. S., Al-Dabbagh, B. S. N., Fadhel, M. A., Manoufali, M., Zhang, J., Al-Timemy, A. H., et al. (2023). A survey on deep learning tools dealing with data scarcity: definitions, challenges, solutions, tips, and applications. *Journal of Big Data*, 10(1):46.

[Aslahishahri et al., 2021] Aslahishahri, M., Stanley, K. G., Duddu, H., Shirtliffe, S., Vail, S., Bett, K., Pozniak, C., and Stavness, I. (2021). From RGB to NIR: Predicting of near infrared reflectance from visible spectrum aerial images of crops. In *2021 IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*, pages 1312–1322.

[Badgley et al., 2017] Badgley, G., Field, C. B., and Berry, J. A. (2017). Canopy near-infrared reflectance and terrestrial photosynthesis. *Science Advances*, 3(3):e1602244.

[Bansal et al., 2022] Bansal, A., Sharma, R., and Kathuria, M. (2022). A systematic review on data scarcity problem in deep learning: Solution and applications. *ACM Computing Surveys*, 54(10s):1–29.

[Davidson et al., 2022] Davidson, C., Jaganathan, V., Sivakumar, A. N., Czarnecki, J. M. P., and Chowdhary, G. (2022). NDVI/NDRE prediction from standard RGB aerial imagery using deep learning. *Computers and Electronics in Agriculture*, 203:107396.

[Dinh et al., 2023] Dinh, B.-D., Nguyen, T.-T., Tran, T.-T., and Pham, V.-T. (2023). 1M parameters are enough? a lightweight CNN-based model for medical image segmentation. In *2023 Asia Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC)*, pages 1279–1284.

[Goodfellow et al., 2014] Goodfellow, I. J., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., and Bengio, Y. (2014). Generative adversarial nets. In *Advances in Neural Information Processing Systems*, volume 27.

[Gutman et al., 2021] Gutman, G., Skakun, S., and Gitelson, A. (2021). Revisiting the use of red and near-infrared reflectances in vegetation studies and numerical climate models. *Science of Remote Sensing*, 4:100025.

[Hossain and Dev, 2026] Hossain, M. Y. and Dev, S. (2026). End-to-end fusion of rgb and low-resolution nir for enhanced crop-specific nir generation. *Computers and Electronics in Agriculture*, 248:111582.

[Hossain et al., 2025] Hossain, M. Y., Rakib, M. M. H., Pakrashi, A., and Dev, S. (2025). Towards efficient spectral translation: A lightweight image-to-image model for rgb-to-nir conversion. In *2025 18th International Congress on Image and Signal Processing, BioMedical Engineering and Informatics (CISP-BMEI)*, pages 1–6.

[Illarionova et al., 2021] Illarionova, S., Shadrin, D., Trekin, A., Ignatiev, V., and Oseledets, I. (2021). Generation of the nir spectral band for satellite images with convolutional neural networks. *Sensors*, 21(16):5646.

[Isola et al., 2017] Isola, P., Zhu, J.-Y., Zhou, T., and Efros, A. A. (2017). Image-to-image translation with conditional adversarial networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5967–5976.

[Jin et al., 2025] Jin, Y., Park, I., Song, H., Ju, H., Nalcakan, Y., and Kim, S. (2025). Pix2Next: Leveraging vision foundation models for RGB to NIR image translation. *Technologies*, 13(4):154.

[Krizhevsky et al., 2012] Krizhevsky, A., Sutskever, I., and Hinton, G. E. (2012). Imagenet classification with deep convolutional neural networks. *Advances in neural information processing systems*, 25.

[Kumar et al., 2024] Kumar, T., Brennan, R., Mileo, A., and Bendechache, M. (2024). Image data augmentation approaches: A comprehensive survey and future directions. *IEEE Access*, 12:187536–187571.

[Minaee et al., 2021] Minaee, S., Boykov, Y., Porikli, F., Plaza, A., Kehtarnavaz, N., and Terzopoulos, D. (2021). Image segmentation using deep learning: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 43(7):1940–1965.

[Picon et al., 2022] Picon, A., Bereciartua-Perez, A., Eguskiza, I., Romero-Rodriguez, J., Jimenez-Ruiz, C. J., Eggers, T., Klukas, C., and Navarra-Mestre, R. (2022). Deep convolutional neural network for damaged vegetation segmentation from RGB images based on virtual NIR-channel estimation. *Artificial Intelligence in Agriculture*, 6:199–210.

[Rebuffi et al., 2021] Rebuffi, S.-A., Gowal, S., Calian, D., Stimberg, F., Wiles, O., and Mann, T. (2021). Data augmentation can improve robustness. In *Proceedings of the 35th International Conference on Neural Information Processing Systems*, NIPS '21, Red Hook, NY, USA. Curran Associates Inc.

[Schmitt et al., 2019] Schmitt, M., Hughes, L. H., Qiu, C., and Zhu, X. X. (2019). SEN12MS – a curated dataset of georeferenced multi-spectral Sentinel-1/2 imagery for deep learning and data fusion. In *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, volume IV-2/W7, pages 153–160.

[Su et al., 2024] Su, J., Luo, Z., Lian, S., Lin, D., and Li, S. (2024). Mutual learning with reliable pseudo label for semi-supervised medical image segmentation. *Medical Image Analysis*, 94:103111.

[van Engelen and Hoos, 2020] van Engelen, J. E. and Hoos, H. H. (2020). A survey on semi-supervised learning. *Machine Learning*, 109(2):373–440.

[Vescovo et al., 2012] Vescovo, L., Wohlfahrt, G., Balzarolo, M., Pilloni, S., Sottocornola, M., Rodeghiero, M., and Gianelle, D. (2012). New spectral vegetation indices based on the near-infrared shoulder wavelengths for remote detection of grassland phytomass. *International Journal of Remote Sensing*, 33(7):2179–2195.

[Xu et al., 2023] Xu, M., Yoon, S., Fuentes, A., and Park, D. S. (2023). A comprehensive survey of image augmentation techniques for deep learning. *Pattern Recognition*, 137:109347.

[Zamir et al., 2023] Zamir, S. W., Arora, A., Khan, S., Hayat, M., Khan, F. S., Yang, M.-H., and Shao, L. (2023). Learning enriched features for fast image restoration and enhancement. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(2):1934–1948.

[Zhao et al., 2024] Zhao, G., He, Y., Wang, Z., Wu, H., and Cheng, L. (2024). Generation of NIR spectral band from RGB image with wavelet domain spectral extrapolation generative adversarial network. *Computers and Electronics in Agriculture*, 227:109461.